Interim Design Report

Micromouse Subsystem



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Chapter 1

Introduction

1.1 Problem Description

Micromouse robotics competitions (originating in the late 1970s) challenge participants to design a robot capable of navigating mazes as quickly and efficiently as possible. It involves integrating hardware modules such as sensing, processing, and power onto a motherboard. This report focuses on the power subsystem, which must distribute power to all other subsystems while adhering to strict constraints. Key challenges include:
Charging a LiPo 800mAh 3.7V battery; Dual-mode charging, higher and lower current charging; Providing two power pin-outs, 3V3 and 5V; Providing an ON/OFF switch; Operating up to four bidirectional motors; Monitoring the battery; Integrating USB C; Driving and switching two external loads

1.2 Scope and Limitations

The project focuses on designing the power subsystem of the Micro-Mouse, specifically on:
Charging a LiPo 800mAh 3.7V battery; Dual-mode charging, higher and lower current charging; Providing two power pin-outs, 3V3 and 5V; Providing an ON/OFF switch; Operating up to four bidirectional motors; Monitoring the battery; Integrating USB C; Driving and switching two external loads

The scope is limited to the design and simulation of the above functionalities, considering the layout for integration with the Micro-Mouse motherboard through a 2x16 pin header and JST battery connector. The dimensions of the PCB are constrained by the maximum dimensions given as 82mm×60mm.

Design decisions and component selections are further constrained by a combined group budget of $70 and approximately $28.25 allocated for components after manufacturing costs. The components selected must be available through JLCPCB with extended parts being charged an additional fee.

1.3 GitHub Link

The design files, schematics, and version history are maintained in the group repository:
https://github.com/PontshoMbizoUCT/MM_power_subsystem/tree/master

Chapter 2

Requirements Analysis

2.1 Requirements

The requirements for the micromouse power module are described in Table 2.1.

Table 2.1: System-level requirements for the power subsystem.

Requirement ID	Description
R01	The system must charge a single-cell 3.7V LiPo battery
R02	The system must support two charging modes: low and high current charging.
R03	The subsystem must output a regulated 3.3V
R04	The subsystem must output a regulated 5V
R05	Include an ON/OFF switch that cuts all power outputs and reduces quiescent currents
R06	The subsystem must drive bidirectional operation of up to four brushed DC motors.
R07	The subsystem must monitor the battery voltage and current
R08	The subsystem must draw 9V from a USB-C input
R09	The subsystem must drive two external loads

2.2 Specifications

The specifications, refined from the requirements in Table 2.1, for the micromouse power module are described in Table 2.2.

Table 2.2: Specifications of the power subsystem derived from the system requirements.

Specification ID	Description
SP01	The battery must be charged by a 9V (HV) constant-voltage
SP02	The value of the lower charging current must be 200mA
SP03	The value of the high charging current and its tolerance must be $600mA \pm 100mA$
SP04	3.3V output must remain at a voltage (and its tolerance) of 3V3±5%
SP05	3.3V output must have a maximum current of $300mA$
SP06	5V output must remain at a regulated voltage (with tolerance) of 5V±5%
SP07	5V output must have a maximum current of $1.5A$
SP08	The minimum allowed quiescent current to be $30\mu A$ under the OFF condition
SP09	The maximum current provided by the battery must be $2A$ in the ON condition

2.3 Acceptance Testing Procedures

Table 2.3: Subsystem acceptance tests

ID	Description	Testing Procedure	Pass/Fail Criteria
AT01	The battery must be charged using a constant 9V	Measure the HV (high voltage) input terminal and battery's temperature using digital thermometer during charging	The 9V must be constant ($9V \pm 0.5V$), and the battery's temperature must be $\leq 60°C$
AT02	Verify lower charging current	Select lower charging mode by setting FAST_CHARGE CTRL to 0V, measure the current in series with the battery using a digital ammeter and monitor the battery's temperature using digital thermometer	The Low charging current must be 200mA±20mA and the battery temperature must be $\leq 60°C$
AT03	Verify higher charging current	Select higher charging mode by setting FAST_CHARGE CTRL to 3.3V, measure the current in series with the battery using a digital ammeter and monitor the battery's temperature using digital thermometer	The High charging current must be 600mA±100mA and the battery temperature must be $\leq 60°C$
AT04	3.3V output voltage	Measure output voltage using a digital voltmeter/multimeter in parallel with the 3.3V pin in no-load (NL) conditions	Voltage must remain between 3.45V and 3.14V in NL conditions
AT05	3.3V maximum output current	Measure output current in series with a varying load [10Ω, 5kΩ] using a digital ammeter	Current must reach a maximum of ≈300mA at low impedance condition (10Ω) and ≈0.7mA at high impedance conditions ($5k\Omega$)
AT06	5V output voltage	Measure output voltage using a digital voltmeter/multimeter in parallel with the 5V pin in no-load (NL) conditions and varying loads [100Ω, 5kΩ]	Voltage remains between 4.75V and 5.25V in NL conditions and varying load conditions
AT07	5V maximum output current	Measure output current in series with a varying load [3.33Ω, 5kΩ] using a digital ammeter	Current must reach a maximum of ≈1.5A at low impedance condition (3.33Ω) and ≈1mA at 5kΩ
AT08	minimum quiescent current (ON/OFF functionality)	Turn OFF switch, measure the series current using a digital ammeter at the battery input while it's not charging.	Current drawn must be $<30\mu$A
AT09	maximum current provided by battery (ON/OFF functionality)	Connect a low impedance load, 10Ω to 3.3V, 5V, all high-side drivers and all motor rails, Turn the switch ON and measure the series current and the battery temperature.	The system must deliver 2A total while the temperature of the battery below $60°C$

Chapter 3

Subsystem Design

3.1 Design Decisions

3.1.1 Design Decision 1: Charging Circuit Selection

Problem: The power subsystem must safely charge a 3.7V 800mAh single-cell LiPo battery from a 9V input with two selectable charging currents (SP01, S02, S03).

The options Considered were:
MCP73831-2-OT: Simple linear charger, supports up to 500mA charging current [1]. **TP4056**: Popular charger with built-in protection, supports up to 1A charging current [2]. **FM4057S52**: Programmable Charge Current: [15mA, 500mA], supports 1.5A [3].

Table 3.1: Comparison of charging ICs

Feature	MCP73831-2-OT	TP4056	FM4057S52
Max Charging Current	500mA	1A	1.5A
Current Selection	Adjustable via Rprog	Adjustable via Rprog	Adjustable via Rprog
Ambient Temperature Range	–40°C to +85°C	0°C to +85°C	–40°C to +85°C
Input Voltage Range	3.75–6V	4.5–8V	4.5–8V
JLCPCB Cost	$1.6815	$0.1719	$0.0545
Extended Component	Yes	Yes	Yes
Availability (JLCPCB)	In stock	Limited stock	Limited stock

Decision: The **MCP73831-2-OT** was selected for its simplicity and easy implementation. The charging current can be varied via a grounding resistor (SP02, SP03). It can integrate well with a USB-C 9V source after stepping down (to 6V) and handles ambient thermals well with a tolerance of 40°C to +85°C, making it reliable in the long term. It's well within budget and has a high stock[1].

3.1.2 Design Decision 2: Voltage Regulator Selection for 3V3 output

Problem: The system requires a regulated 3.3V output with ±5% accuracy (SP04) capable of supplying up to 300mA (SP05).

The options Considered were:
AMS1117-3.3: low-cost, simple 3V3 linear regulator [4]. **MIC5205-3.3YM5**: A low dropout linear regulator with excellent noise performance and a low quiescent current [5]. **AP2112K-3.3TRG1**: A low-dropout linear regulator with low quiescent current and stable 3.3V fixed output; suitable for powering low-power ICs [6].

Table 3.2: Comparison of 3.3V regulator option

Feature	AMS1117-3.3	MIC5205-3.3YM5	AP2112K-3.3
Type	Linear Regulator	LD Regulator	LD Linear Regulator
Max Load Regulation	0.4%/A	0.5%	1%/A
Max Output Current	1A	150mA	600mA
Max Dropout Voltage	1.3V	0.350V	1.3V
Max Quiescent Current	11mA	5μA	55μA
Operating Junction Temperature	-40 to 125°C	-40 to 125°C	150°C
Output Noise	$99mV_{RMS}$	$80\mu V_{RMS}$	$50\mu V_{RMS}$
cost	$0.2144	$0.3501	$0.0729
Extended	yes	yes	yes
JLCPCB Availability	In stock	Limited Stock	In stock

Decision: The **AP2112K-3.3TRG1** was selected because of its low quiescent current of 55μA and a large junction temperature of 150°C. It provides a constant regulated 3V3 voltage and a maximum output current of 600mA (SP04, SP05). It also has excellent output stability with a load regulation of 1%/A and low dropout voltage of 1.3V. It is cheap and has high stock. It balances performance (3.2), cost, and availability, meeting the defined specifications [6].

3.1.3 Design Decision 3: Booster Circuit Selection for 5V Output

Problem: The system requires a constant, regulated 5V output with ±5% tolerance (SP06) capable of supplying up to 1.5A (SP07).

The options considered were:
MT3608: A simple, low-cost boost converter with up to 2A switch current [7]. **MC34063ADR2G**: A regulator that can be configured for boost or buck operation [8]. **TPS61220DCKR**: A synchronous boost converter [9].

Table 3.3: Comparison of 5V booster circuits

Feature	MT3608	MC34063ADR2G	TPS61220DCKR
Type	Boost Converter	Buck/Boost Converter	Synchronous Boost Converter
Max Output Current	2A	1.5A	200mA @ 5V
Input Voltage	[2V,24V]	[3V,40V]	[-0.3V,5.5V]
Max Output Voltage	28V	40V	6V
Max Efficiency	97%	88% (step-up)	95%
Quiescent Current	200μA	100μA	5.5μA
Cost (JLCPCB)	$0.0599	$0.4920	$0.3720
Extended Component	Yes	Preferred	Yes
JLCPCB Availability	In stock	Limited stock	Limited stock

Decision: The **MT3608** was selected because it meets the current specifications of 1.5A (SP07). It has a high efficiency of 97% and its 2A continuous output capability makes it well-suited for supplying the regulated 5V output (SP06). The MT3608 meets the specifications and the thermal and budget constraints [7].

3.2 Final Design

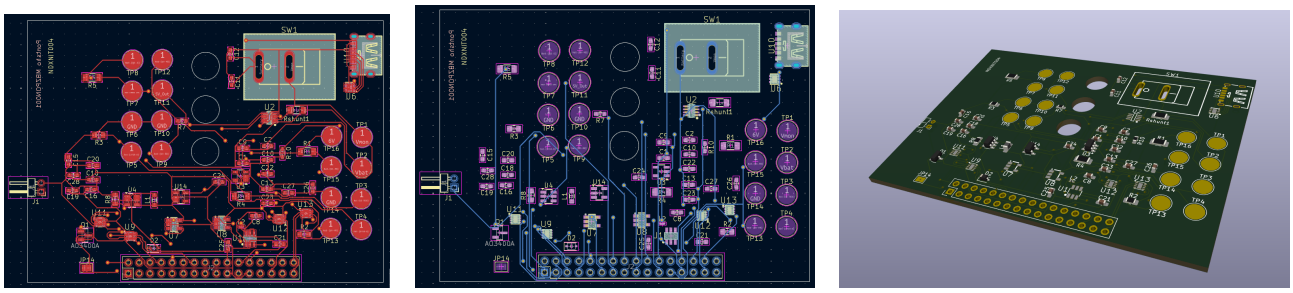
The final design meets all requirements: dual-mode charging, regulated 3.3V and 5V outputs, and an ON/OFF switch—developed within strict electrical and budget constraints.

The **MCP73831-2-OT** requires a 6V input to charge the battery, while the system supplies 9V (SP01). A voltage divider was considered but replaced with the **MT2492** buck converter because of its regulated output and high efficiency [1][10]. It accepts the 9V input and outputs 6V at up to 2A, and is labeled U1 on the schematic (3.2).

To enable digital control of the charging current (SP02, SP03), an N-channel E-MOSFET was used (compared to a BJT) to switch between two R_{PROG} resistors. This approach provides consistent performance due to the MOSFET's low and stable on-resistance and negligible gate current, avoiding the inefficiencies and variability of a BJT.

For ON/OFF control (R05), a mechanical STSP switch was placed between the system power line and the battery output, allowing power to the system to be disconnected while keeping the battery manager connected to the raw battery voltage for charging. This maintains charging functionality even when the system is OFF.

The **AP2112K-3.3** was selected for 3.3V (SP04) regulation due to its low quiescent current and cost-effectiveness [4], while the **MT3608** boost converter was chosen for its efficiency and ability to meet the 1.5A 5V output (SP06,SP07).



(a) Front PCB (b) Back PCB (c) 3D PCB

Figure 3.1: Final PCB layout with labeled layers and 3D view.

Table 3.4: Bill of Materials (BOM)

Comment	**Designator(s)**	**Footprint**	**LCSC Part Number**	**Qty**
4.7uH	L2	L_0201_0603Metric Pad0.64x0.40mm Hand-Solder	C1034	1
MCP73831-2-OT	U3	SOT-23-5	C14879	1
11kΩ	R2	R_0805_2012Metric Pad1.20x1.40mm Hand-Solder	C17429	1
1N5819WS	D2	D_SOD-323	C191023	1
15kΩ	R8	R_0805_2012Metric Pad1.20x1.40mm Hand-Solder	C17475	1
3kΩ	R3	R_0805_2012Metric Pad1.20x1.40mm Hand-Solder	C17661	1
100kΩ	R1	R_1206_3216Metric Pad1.30x1.75mm Hand-Solder	C17900	1
10kΩ	R5	R_1206_3216Metric Pad1.30x1.75mm Hand-Solder	C17902	1
2kΩ	R4	R_1206_3216Metric Pad1.30x1.75mm Hand-Solder	C17944	1
4.7uF	C1, C2	C_0603_1608Metric Pad1.08x0.95mm Hand-Solder	C19666	2
110kΩ	R7	R_0603_1608Metric Pad0.98x0.95mm Hand-Solder	C25805	1
22uH	L1	L_0805_2012Metric Pad1.05x1.20mm Hand-Solder	C32375	1
22uF	C21, C22, C23, C24, C25, C8, C9	C_0603_1608Metric Pad1.08x0.95mm Hand-Solder	C59461	7
MT3608	U4	SOT-23-6	C84817	1
MT2492	U1	MT2492	C89358	1
AP2112K-3.3	U14	SOT-23-5	—	1

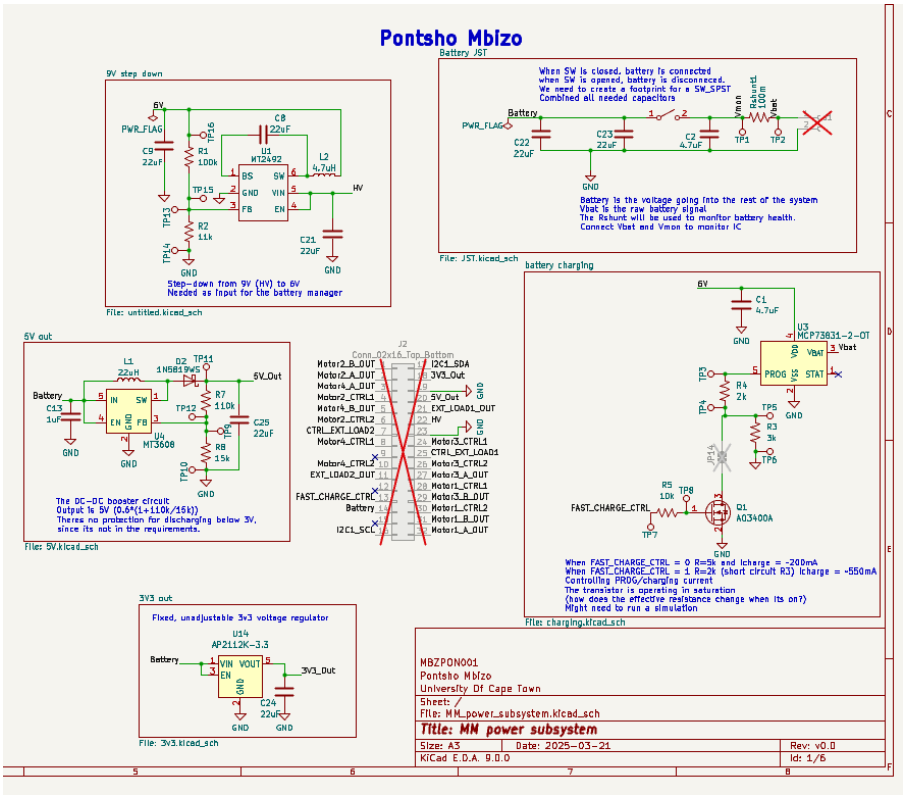


Figure 3.2: Final schematic with labeled subsystems and annotated components.

3.3 Failure Management

The following table highlights measures implemented in the power subsystem to correct/isolate potential failures after manufacturing.

Table 3.5: Failure mitigation strategies in power subsystem

Name	Description
Test Points for Key Nodes	Test points were added on important power_outputs-IC_inputs to allow for easy multimeter debugging.
The use of SOT_SMD package	The SOT_SMD footprint package was chosen because its exposed leads allows the soldering of test wires or manually apply voltages.
Jumper Pads on selecting logic circuits	Pads were placed to allow easy removal of the charging logic circuit and manual selection between fast and slow charge modes in case of failure.

3.4 System Integration and Interfacing

To integrate the power subsystem with the micromouse motherboard, the PCB was designed to connect directly to a 2x16 header with I/O pins to the motherboard, battery, motors, and the external loads.

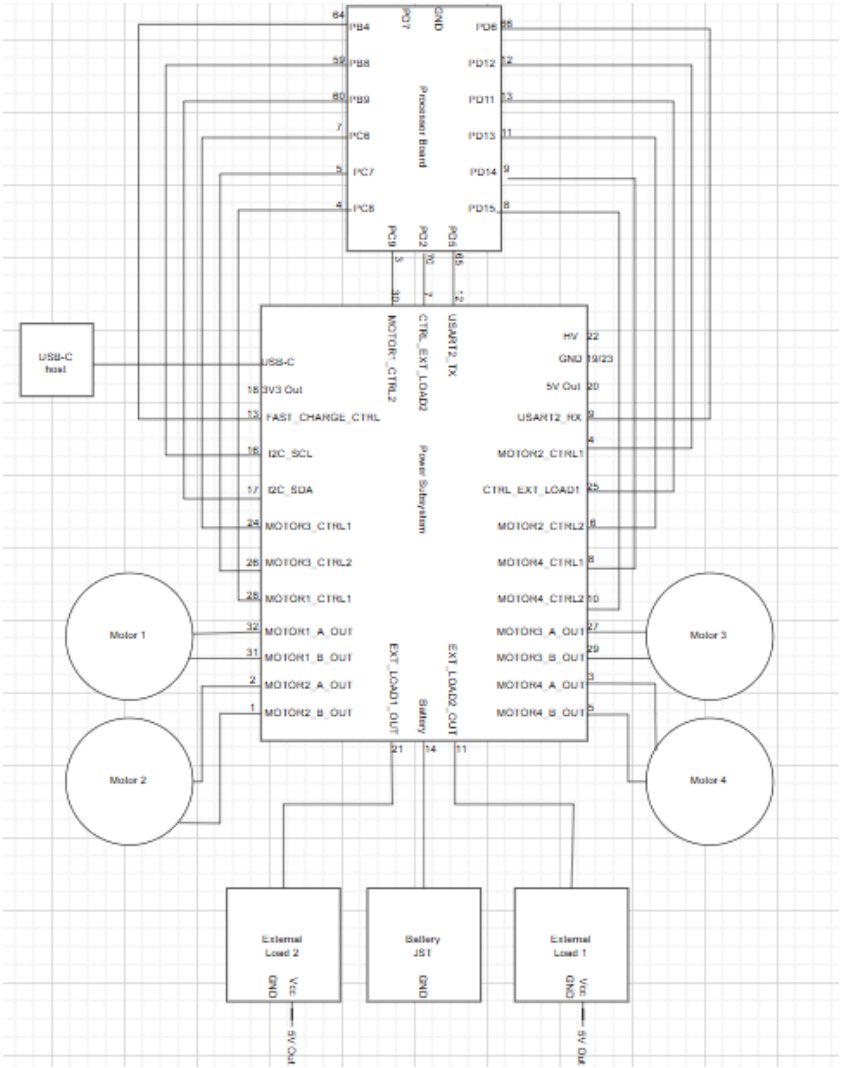


Figure 3.3: system integration diagram

Table 3.6: Interfacing specifications

Interface	Description	Pins/Output
I001	I/O connections between the power subsystem and the Processor board	• CTRL_EXT_LOAD1 is connected to pin PD11 on the Processor board • CTRL_EXT_LOAD2 is connected to pin PD2 on the Processor board • USART2_RX is connected to pin PD15 on the Processor board • USART2_TX is connected to pin PD5 on the Processor board • FAST_CHARGE_CTRL is connected to pin PB4 on the Processor board • I2C1_SCL is connected to pin PB8 on the Processor board • I2C1_SDA is connected to pin PB9 on the Processor board • MOTOR1_CTRL1 is connected to pin PC8/TIM3C3 on the Processor board • MOTOR1_CTRL2 is connected to pin PC9/TIM3C4 on the Processor board • MOTOR2_CTRL1 is connected to pin PD12/TIM4C1 on the Processor board • MOTOR2_CTRL2 is connected to pin PD13/TIM4C2 on the Processor board • MOTOR3_CTRL1 is connected to pin PC6/TIM3C1 on the Processor board • MOTOR3_CTRL2 is connected to pin PC7/TIM3C2 on the Processor board • MOTOR4_CTRL1 is connected to pin PD14/TIM4C3 on the Processor board • MOTOR4_CTRL2 is connected to pin PD15/TIM4C4 on the Processor board
I002	Connections between the power subsystem and the LiPo battery	• Battery is connected to the Lipo battery via a JST connector
I003	Connections between the power subsystem and the bidirectional motors	• Motor1_A_OUT,Motor1_B_OUT are connected to Motor1 • Motor2_A_OUT,Motor2_B_OUT are connected to Motor2 • Motor3_A_OUT,Motor3_B_OUT are connected to Motor3 • Motor4_A_OUT,Motor4_B_OUT are connected to Motor4
I004	Connections between the power subsystem and the external loads	• EXT_LOAD1_OUT is connected to Load1 • EXT_LOAD1_OUT is connected to Load1

Chapter 4

Discussion and Results

The power subsystem was successfully manufactured then tested in the lab. Each acceptance test was performed using a digital multimeter and specific loads. The results of the acceptance tests are discussed below.

AT01 – The battery must be charged using a constant 9V: The HV output remained constant at 9.05V when the USB was plugged in. The step-down converter (U1, MT2429) produced a constant 6.025V output that was fed into the charging IC (U3, MCP73831-2-OT). The charging IC produced 4.15V at its output, which is used to charge the battery, confirming proper operation of the charging path. The measurements were captured using a multimeter. The ATP passed because the measured values aligned with design expectations for the charging voltages.

AT02 – Low Current Charging Mode (200mA): To verify charging functionality, a simulated battery setup was used. Two resistors were connected in series: $R_{S1} = 1\Omega$, connected to the charging IC (U3, MCP73831-2-OT) output, and $R_{S2} = \frac{3.2V}{200mA} \approx 15\Omega$, used to set the charging current. The junction between R_{S1} and R_{S2} was connected to a programmable power supply (PSU) set to 3.2V to simulate a drained battery.

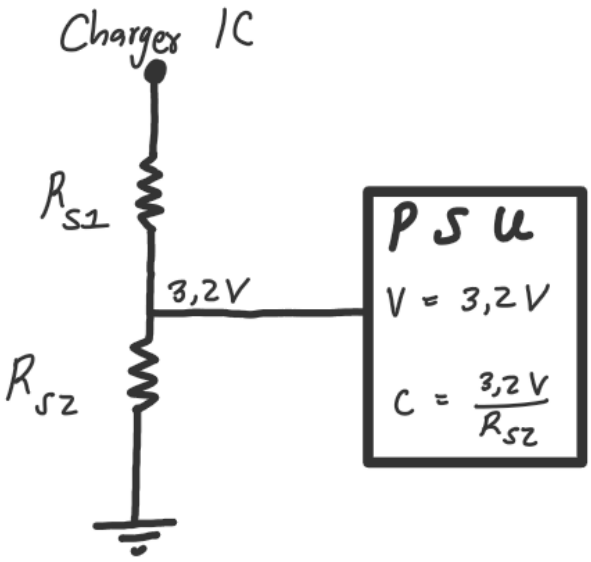


Figure 4.1: Simulated battery used for charging ATP.

Initially (USB-C not connected), the current supplied by the PSU read 215mA. When USB-C power was applied, the PSU current dropped to 2mA, and the current through R_{S2} remained constant at

$\approx 202mA$. This confirmed that the charging IC (U3, MCP73831-2-OT) was sourcing current into the simulated battery. The ATP passed because the charging IC was able to source 200mA into a simulated battery (SP02).

AT03 – High Current Charging Mode (600mA ±100mA target): The same simulated battery was used with the control signal (FAST_CHARGE_CTRL) set HIGH (3.3V). The measured impedance between TP03 and TP06 (between the resistors, R3 and R4) dropped from $5k\Omega$ to $2k\Omega$, confirming that the charging control circuit works. $R_{PROG} = 2k\Omega$ was intended to increase the charging current to $\approx 500mA$. R_{S2} was changed to $10\Omega \parallel 10\Omega = 5\Omega$.

Initially (USB-C not connected) The measured current was 520mA (from the PSU). When USB-C power was applied, the PSU current dropped to 200mA, and the current through R_{S2} measured at 330mA, significantly below the expected 600mA ±100mA range. The ATP failed because the charging IC was unable to source 500mA into a simulated battery (SP03). This could've been caused by the output current being restriced by the PCB tracks used (0.25mm tracks) and using an IC that could supply a maximum of 500mA.

AT04 – 3.3V output voltage: When the battery [3.7V-4.2V] was connected and the mechanical switch was ON: a constant 3.31V, measured using a multimeter, was produced at the output of the 3V3 regulator (U14, AP2112K-3.3) . The ATP passed because the output remained within the ±5% tolerance of 3V3 specified by SP04.

AT05 – 3.3V maximum output current: Current through the 3.3V rail was measured under a 10Ω and $5k\Omega$ load using a multimeter connnected in series with the loads. For the 10Ω load, the measured current was 327mA and 0.75mA when $5k\Omega$ was connected. The ATP passed because the current draw under load reached 300mA and demonstrated compliance with SP05.

AT06 – 5V output voltage: A constant 5.05V, measured using a multimeter, was produced at the output of (U4, MT3608) when the battery [3.7V-4.2V] was connected and the switch was ON. The ATP passed because the output remained within the ±5% tolerance of 5V specified by SP06.

AT07 – 5V maximum output current: Current through the 5V rail was measured under a 3.33Ω (which is was made by connecting 3 10Ω in parallel) and $5k\Omega$ load using a multimeter connnected in series with the loads. For the 3.33Ω load, the measured current was 1A and 1mA when $5k\Omega$ was connected. The ATP failed because the specification required delivery of up to 1.5A (SP07) This could be caused by the output current being restriced by the PCB tracks used (0.25mm tracks). This could be improved by using power tracks of width (1.2mm) to allow the better flow of current.

In summary, most acceptance tests passed except for higher current specifications (SP03, SP07, SP09). The power subsystem meets its low charging mode and voltage regulation requirements, but fails the higher current specifications; The value of the high charging current and its tolerance must be $600mA \pm 100mA$ (SP03), 5V output must have a maximum current of 1.5A (SP07), The maximum current provided by the battery must be 2A in the ON condition (SP09). This could be a result of using signal tracks instead of power tracks for the 5V, battery, 9V tracks.

Chapter 5

Conclusion

The aim is to present the design, implementation, and testing of a power subsystem for the micromouse, developed as part of the EEE3088F 2025 course. The subsystem is required to perform multiple functions; this includes safely charging a LiPo battery, dual-mode current selection, voltage regulation at 3.3V and 5V, and switching power ON/OFF, all designed within electrical, mechanical, and financial constraints.

The final solution featured a MCP73831 charging circuit with selectable 200mA and $600mA \pm 100mA$ charging currents, a 3.3V regulator using the AP2112K, and a 5V, MT3608 boost converter. The PCB was manufactured then tested in the lab.

The power subsystem met most of the specified requirements, including low-current battery charging, regulated 3.3V and 5V outputs, and ON/OFF power control. The low charging mode (200mA), voltage regulation for both rails, and quiescent current performance were all tested and matched expected behavior under different loads.

The subsystem failed to meet higher current specifications. The high-current charging mode (SP03) did not reach the required $600mA \pm 100mA$, likely due to the limitations of the MCP73831-2-OT IC (500mA maximum output) and narrow PCB traces restricting current flow. The 5V rail failed to deliver the specified maximum of 1.5A (SP07), and the system could not supply the full 2A output in the ON state (SP09). These failures are because of the small track widths (0.25mm) used for high-current paths, such as battery, 5V, and HV.

5.1 Recommendations

Recommendations:

- Increasing PCB track widths to 1.2mm on all high-current paths (Battery, HV, 5V) to reduce resistive losses and prevent voltage drops along the tracks.
- Replacing the MCP73831-2-OT with a battery management IC like TP4056 to reliably support the 600mA fast charging mode.
- Adding thermal reliefs on the high-current paths (Battery, HV, 5V) to improve heat dissipation and allow high current operation.

Chapter 6

Reflections

Design Phase: Too much time was spent looking at ICs. Early selection of different types of ICs followed by evaluation would save time.

Board Layout: Searching for the correct footprints caused unnecessary delays. The use of EasyEDA's library JLCPCB should've been the first option. The use of Autorouter was slower than anticipated; manually routing the PCB would've been more efficient and would've allowed us to include power tracks„ which caused failures in the ATP.

Post-Facto Gantt Chart:

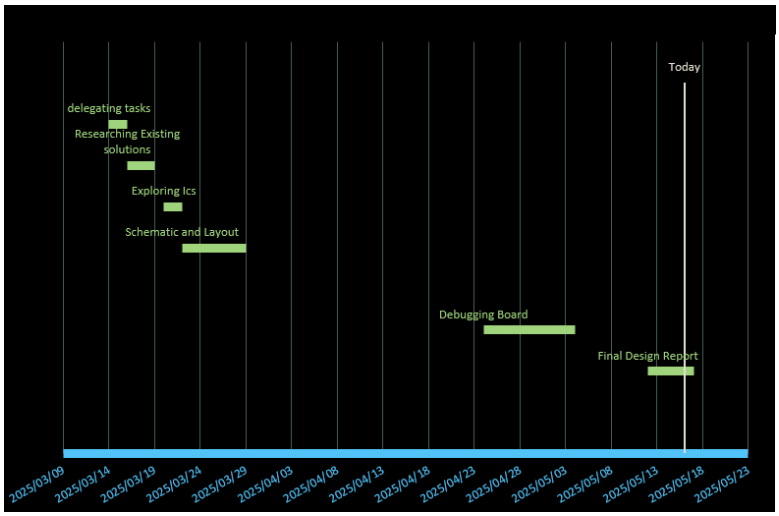


Figure 6.1: post-facto Gantt chart of time spent on each task.

Cost Reduction:
Passive components could've been used for the external load switches instead of an IC, which costs more.

Motor Control: If two motors differ in characteristics, direct PWM control will result in uneven speeds, causing the micromouse to drift off-course. Implementing a feedback system using motor encoders and a PID controller to adjust the PWM based on the difference between the desired and measured motor speeds will ensure consistent speeds.

Motor faults: The H-bridge includes thermal shutdown and current sensing; this will protect the bridge in case of any faults (shorting and stalling).

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